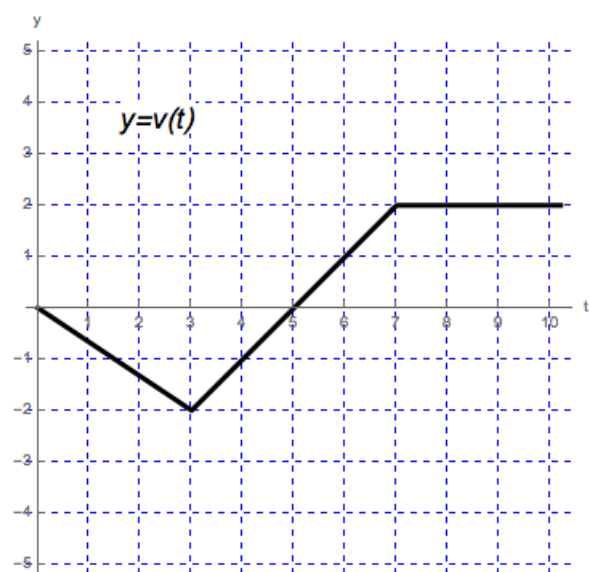


Problem 1

Problem 1

Problem 1 The graph of the velocity (t in min, v in ft/min) of a particle moving along a straight line is given in the figure. Assume that the particle was at the origin initially.



(a) Evaluate the displacement of the particle over the following intervals.

(i) $[0, 10]$

Explanation.

$$s(10) - s(0) = \int_0^{10} v(t) dt = 3ft$$

(ii) $[0, 7]$

Explanation.

$$s(7) - s(0) = -3ft$$

(iii) $[0, 5]$

Explanation.

$$s(5) - s(0) = -5ft$$

Problem 1

- (b) *When was the particle farthest from the origin? [3]*

Explanation. We check the end points: $s(0) = 0$, $s(10) = s(0) + 3 = 3$, and critical points. Since the only critical point of s is at $t = 5$, since $s'(5) = v(5) = 0$ there, we evaluate $s(5) = s(0) - 5 = -5$. It follows that the particle was farthest from the origin at $t = 5$ min.

- (c) *What was the total distance the particle has travelled over the time interval $[0, 10]$?*

Explanation. The total distance travelled is given by the definite integral

$$\int_0^{10} |v(t)| dt = \int_0^5 (-v(t)) dt + \int_5^{10} v(t) dt = 5 + 8 = 13 ft$$